

Labsheet-2

1. Write a Java program to find the area of a triangle given its three sides using Heron's formula.

```
import java.util.Scanner;
```

```
public class TriangleArea {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input three sides
```

```
        System.out.print("Enter side a: ");
```

```
        double a = sc.nextDouble();
```

```
        System.out.print("Enter side b: ");
```

```
        double b = sc.nextDouble();
```

```
        System.out.print("Enter side c: ");
```

```
        double c = sc.nextDouble();
```

```
        // Calculate semi-perimeter
```

```
        double s = (a + b + c) / 2;
```

```
        // Heron's formula
```

```
        double area = Math.sqrt(s * (s - a) * (s - b) * (s - c));
```

```
        // Display area
```

```
System.out.println("Area of the triangle = " + area);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter side a: 5

Enter side b: 6

Enter side c: 7

Area of the triangle = 14.696938456699069

2. Create a Java program to calculate the compound interest given principal, rate, and time.

```
import java.util.Scanner;
```

```
public class CompoundInterest {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input principal, rate and time
```

```
        System.out.print("Enter Principal amount: ");
```

```
        double p = sc.nextDouble();
```

```
        System.out.print("Enter Rate of interest: ");
```

```
        double r = sc.nextDouble();
```

```
        System.out.print("Enter Time (in years): ");
```

```
        double t = sc.nextDouble();
```

```

// Calculate amount
double amount = p * Math.pow((1 + r / 100), t);

// Calculate compound interest
double ci = amount - p;

System.out.println("Compound Interest = " + ci);
System.out.println("Total Amount = " + amount);

sc.close();
}
}

```

OUTPUT:-

Enter Principal amount: 10000

Enter Rate of interest: 5

Enter Time (in years): 2

Compound Interest = 1025.0

Total Amount = 11025.0

3. Develop a Java program to find the distance between two points in a 2D plane using the distance formula.

```
import java.util.Scanner;
```

```

public class DistanceBetweenPoints {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

```

```
// Input coordinates of first point
System.out.print("Enter x1: ");
double x1 = sc.nextDouble();

System.out.print("Enter y1: ");
double y1 = sc.nextDouble();

// Input coordinates of second point
System.out.print("Enter x2: ");
double x2 = sc.nextDouble();

System.out.print("Enter y2: ");
double y2 = sc.nextDouble();

// Calculate distance
double distance = Math.sqrt(
    Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2)
);

System.out.println("Distance between the two points = " + distance);

sc.close();
}
}
```

OUTPUT:-

Enter x1: 2

Enter y1: 3

Enter x2: 5

Enter y2: 7

Distance between the two points = 5.0

4. Write a Java program to demonstrate prefix and postfix increment by simulating a counter for visitors entering and leaving a store.

```
public class VisitorCounter {  
    public static void main(String[] args) {  
  
        int visitors = 0;  
  
        // Postfix increment: first use the value, then increment  
        System.out.println("Visitors before entry: " + visitors);  
        System.out.println("Visitor entering (postfix): " + visitors++);  
        System.out.println("Visitors after entry: " + visitors);  
  
        // Prefix increment: first increment, then use the value  
        System.out.println("\nVisitor entering (prefix): " + ++visitors);  
        System.out.println("Total visitors: " + visitors);  
  
        // Simulating a visitor leaving  
        System.out.println("\nVisitor leaving...");  
        visitors--;  
  
        System.out.println("Current visitors in store: " + visitors);  
    }  
}
```

```
}  
}
```

OUTPUT:-

Visitors before entry: 0

Visitor entering (postfix): 0

Visitors after entry: 1

Visitor entering (prefix): 2

Total visitors: 2

Visitor leaving...

Current visitors in store: 1

5. Create a Java program that takes an integer and outputs its negative value using unary operators.

```
import java.util.Scanner;
```

```
public class NegativeValue {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter an integer: ");  
        int num = sc.nextInt();  
  
        // Unary minus operator  
        int negative = -num;
```

```
System.out.println("Negative value = " + negative);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter an integer: 25

Negative value = -25

6. Write a Java program to gradually reduce a number by half using /= until it becomes less than 1, counting the number of steps.

```
import java.util.Scanner;
```

```
public class HalfNumber {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
```

```
        double num = sc.nextDouble();
```

```
        int steps = 0;
```

```
        // Reduce the number by half until it becomes less than 1
```

```
        while (num >= 1) {
```

```
            num /= 2;
```

```
            steps++;
```

```
        }
```

```
System.out.println("Final value = " + num);  
System.out.println("Number of steps = " + steps);
```

```
sc.close();  
}  
}
```

OUTPUT:-

Enter a number: 16

Final value = 0.5

Number of steps = 5

7. Create a Java program to accumulate rainfall data for 7 days using += and find the total.

```
import java.util.Scanner;
```

```
public class Rainfall {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        double totalRainfall = 0;  
  
        // Enter rainfall for 7 days  
        for (int day = 1; day <= 7; day++) {  
            System.out.print("Enter rainfall for day " + day + " (in mm): ");  
            double rainfall = sc.nextDouble();  
  
            totalRainfall += rainfall;
```

```
}
```

```
System.out.println("Total rainfall for 7 days = " + totalRainfall + " mm");
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter rainfall for day 1 (in mm): 10

Enter rainfall for day 2 (in mm): 5

Enter rainfall for day 3 (in mm): 8

Enter rainfall for day 4 (in mm): 12

Enter rainfall for day 5 (in mm): 6

Enter rainfall for day 6 (in mm): 9

Enter rainfall for day 7 (in mm): 10

Total rainfall for 7 days = 60.0 mm

8. Write a Java program to check whether three given angles form a valid triangle.

```
import java.util.Scanner;
```

```
public class ValidTriangle {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter first angle: ");
```

```
        int angle1 = sc.nextInt();
```

```

System.out.print("Enter second angle: ");

int angle2 = sc.nextInt();


System.out.print("Enter third angle: ");

int angle3 = sc.nextInt();


// Check validity of triangle
if (angle1 > 0 && angle2 > 0 && angle3 > 0
    && angle1 + angle2 + angle3 == 180) {
    System.out.println("The angles form a valid triangle.");
} else {
    System.out.println("The angles do not form a valid triangle.");
}


sc.close();
}
}

```

OUTPUT:-

Enter first angle: 60

Enter second angle: 60

Enter third angle: 60

The angles form a valid triangle.

9. Create a Java program to compare two strings for lexicographic order without using built-in compare functions (use relational operators on characters).

```
import java.util.Scanner;
```

```
public class LexicographicCompare {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter first string: ");  
        String str1 = sc.nextLine();  
  
        System.out.print("Enter second string: ");  
        String str2 = sc.nextLine();  
  
        int minLength = Math.min(str1.length(), str2.length());  
        int result = 0;  
  
        // Compare characters one by one  
        for (int i = 0; i < minLength; i++) {  
            char ch1 = str1.charAt(i);  
            char ch2 = str2.charAt(i);  
  
            if (ch1 < ch2) {  
                result = -1;  
                break;  
            } else if (ch1 > ch2) {  
                result = 1;  
                break;  
            }  
        }  
    }  
}
```

```
}
```

```
// If all common characters are equal, compare lengths
```

```
if (result == 0) {
```

```
    if (str1.length() < str2.length()) {
```

```
        result = -1;
```

```
    } else if (str1.length() > str2.length()) {
```

```
        result = 1;
```

```
    }
```

```
}
```

```
if (result == 0) {
```

```
    System.out.println("Both strings are equal.");
```

```
} else if (result < 0) {
```

```
    System.out.println("First string comes before second string.");
```

```
} else {
```

```
    System.out.println("First string comes after second string.");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter first string: Apple

Enter second string: Banana

First string comes before second string.

10. Write a Java program to determine if a given year is both a leap year and within a given range.

```
import java.util.Scanner;
```

```
public class LeapYearRange {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter year: ");
```

```
        int year = sc.nextInt();
```

```
        System.out.print("Enter lower limit: ");
```

```
        int lower = sc.nextInt();
```

```
        System.out.print("Enter upper limit: ");
```

```
        int upper = sc.nextInt();
```

```
        // Check leap year
```

```
        boolean leapYear = (year % 400 == 0) ||
```

```
            (year % 4 == 0 && year % 100 != 0);
```

```
        // Check if year is within range
```

```
        boolean withinRange = (year >= lower && year <= upper);
```

```
        if (leapYear && withinRange) {
```

```
            System.out.println(year + " is a leap year and is within the given range.");
```

```
    } else {  
        System.out.println(year + " does not satisfy both conditions.");  
    }  
  
    sc.close();  
}  
}
```

OUTPUT:-

Enter year: 2024

Enter lower limit: 2000

Enter upper limit: 2030

2024 is a leap year and is within the given range.

11. Create a Java program to check if a student passes a course, where they must have at least 40% in theory AND 50% in practical OR overall, 50%.

```
import java.util.Scanner;
```

```
public class CourseResult {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter Theory marks (%): ");  
        double theory = sc.nextDouble();  
  
        System.out.print("Enter Practical marks (%): ");  
        double practical = sc.nextDouble();
```

```

System.out.print("Enter Overall marks (%): ");

double overall = sc.nextDouble();

// Pass condition:

// Theory >= 40% AND Practical >= 50%

// OR Overall >= 50%

if ((theory >= 40 && practical >= 50) || overall >= 50) {

    System.out.println("Student passes the course.");

} else {

    System.out.println("Student fails the course.");

}

sc.close();

}

}

```

OUTPUT:-

Enter Theory marks (%): 45

Enter Practical marks (%): 55

Enter Overall marks (%): 48

Student passes the course.

12. Write a Java program to find the smallest of four numbers using nested ternary operators.

```
import java.util.Scanner;
```

```

public class SmallestOfFour {

    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter four numbers: ");
```

```
int a = sc.nextInt();
```

```
int b = sc.nextInt();
```

```
int c = sc.nextInt();
```

```
int d = sc.nextInt();
```

```
int smallest = (a < b)
```

```
    ? ((a < c) ? ((a < d) ? a : d) : ((c < d) ? c : d))  
    : ((b < c) ? ((b < d) ? b : d) : ((c < d) ? c : d));
```

```
System.out.println("Smallest number = " + smallest);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter four numbers: 25 12 30 8

Smallest number = 8

13. Create a Java program to check if a character is a vowel, consonant, digit, or special symbol using the ternary operator.

```
import java.util.Scanner;
```

```
public class CharacterCheck {
```

```
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter a character: ");
```

```
char ch = sc.next().charAt(0);
```

```
String result = (ch >= '0' && ch <= '9') ? "Digit" :
```

```
((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) ?
```

```
((ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U' ||
```

```
ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
```

```
? "Vowel" : "Consonant")
```

```
: "Special Symbol";
```

```
System.out.println("The character is a " + result);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter a character: A

The character is a Vowel

14. Write a Java program to swap two integers without using a temporary variable (use XOR).

```
import java.util.Scanner;
```

```
public class SwapUsingXOR {
```

```
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter first integer: ");
```

```
int a = sc.nextInt();
```

```
System.out.print("Enter second integer: ");
```

```
int b = sc.nextInt();
```

```
System.out.println("Before swapping: a = " + a + ", b = " + b);
```

```
// Swapping using XOR
```

```
a = a ^ b;
```

```
b = a ^ b;
```

```
a = a ^ b;
```

```
System.out.println("After swapping: a = " + a + ", b = " + b);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter first integer: 10

Enter second integer: 20

Before swapping: a = 10, b = 20

After swapping: a = 20, b = 10.

15. Create a Java program to count the number of set bits (1s) in the binary representation of a number using bitwise operators.

```
import java.util.Scanner;

public class CountSetBits {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int n = sc.nextInt();

        int count = 0;

        int num = n;

        while (num > 0) {

            count = count + (num & 1);

            num = num >> 1;

        }

        System.out.println("Number of set bits = " + count);

        sc.close();

    }

}
```

OUTPUT:-

Enter a number: 13

Number of set bits = 3

16. Write a Java program to perform fast multiplication and division of a number by powers of two using shift operators.

```
import java.util.Scanner;
```

```
public class ShiftOperations {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
```

```
        int num = sc.nextInt();
```

```
        System.out.print("Enter the power of 2: ");
```

```
        int power = sc.nextInt();
```

```
        // Multiplication by 2^power
```

```
        int multiplication = num << power;
```

```
        // Division by 2^power
```

```
        int division = num >> power;
```

```
        System.out.println("After multiplication by 2^" + power + " = " + multiplication);
```

```
        System.out.println("After division by 2^" + power + " = " + division);
```

```
        sc.close();
```

```
    }
```

```
}
```

OUTPUT:-

Enter a number: 20

Enter the power of 2: 2

After multiplication by $2^2 = 80$

After division by $2^2 = 5$

17. Create a Java program to cyclically rotate the bits of an integer to the left by 2 positions.

```
import java.util.Scanner;
```

```
public class RotateBitsLeft {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter an integer: ");
```

```
        int num = sc.nextInt();
```

```
        // Cyclic left rotation by 2 positions
```

```
        int result = (num << 2) | (num >>> 30);
```

```
        System.out.println("Original number = " + num);
```

```
        System.out.println("After cyclic left rotation by 2 bits = " + result);
```

```
        sc.close();
```

```
    }
```

```
}
```

OUTPUT:-

Enter an integer: 5

Original number = 5

After cyclic left rotation by 2 bits = 20

18. Write a Java program to print the first 20 terms of the Fibonacci sequence using a for loop.

```
public class Fibonacci {  
    public static void main(String[] args) {  
  
        int first = 0;  
        int second = 1;  
  
        System.out.println("First 20 terms of Fibonacci sequence:");  
  
        for (int i = 1; i <= 20; i++) {  
            System.out.print(first + " ");  
  
            int next = first + second;  
            first = second;  
            second = next;  
        }  
    }  
}
```

OUTPUT:-

First 20 terms of Fibonacci sequence:

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

19. Create a Java program to print all Armstrong numbers between 1 and 1000 using a for loop.

```
public class ArmstrongNumbers {  
    public static void main(String[] args) {  
  
        System.out.println("Armstrong numbers between 1 and 1000:");  
  
        for (int num = 1; num <= 1000; num++) {  
            int original = num;  
            int sum = 0;  
  
            while (original > 0) {  
                int digit = original % 10;  
                sum = sum + (digit * digit * digit);  
                original = original / 10;  
            }  
  
            if (sum == num) {  
                System.out.print(num + " ");  
            }  
        }  
    }  
}
```

OUTPUT:-

Armstrong numbers between 1 and 1000:

1 153 370 371 407 1000

20. Write a Java program to repeatedly accept a password until the correct one is entered (use do-while).

```
import java.util.Scanner;

public class PasswordCheck {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String correctPassword = "java123";
        String password;

        do {

            System.out.print("Enter password: ");

            password = sc.nextLine();

            if (!password.equals(correctPassword)) {

                System.out.println("Incorrect password. Try again.");

            }

        } while (!password.equals(correctPassword));

        System.out.println("Correct password! Access granted.");

        sc.close();

    }

}
```

OUTPUT:-

Enter password: abc123

Incorrect password. Try again.

Enter password: test

Incorrect password. Try again.

Enter password: java123

Correct password! Access granted.

21. Create a Java program to display all factors of a number using a do-while loop.

```
import java.util.Scanner;
```

```
public class Factors {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter a number: ");  
        int num = sc.nextInt();  
  
        int i = 1;  
  
        System.out.println("Factors of " + num + " are:");  
  
        do {  
            if (num % i == 0) {  
                System.out.print(i + " ");  
            }  
            i++;  
        }  
    }  
}
```

```
    } while (i <= num);

    sc.close();
}
}
```

OUTPUT:-

Enter a number: 12

Factors of 12 are:

1 2 3 4 6 12

22. Write a Java program to reverse the digits of an integer using a while loop.

```
import java.util.Scanner;

public class ReverseNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");

        int num = sc.nextInt();

        int reverse = 0;

        while (num != 0) {

            int digit = num % 10;

            reverse = reverse * 10 + digit;

            num = num / 10;

        }

    }

}
```

```
System.out.println("Reversed number = " + reverse);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter an integer: 12345

Reversed number = 54321

23. Create a Java program to determine if a number is a palindrome using a while loop.

```
import java.util.Scanner;
```

```
public class PalindromeNumber {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
```

```
        int num = sc.nextInt();
```

```
        int original = num;
```

```
        int reverse = 0;
```

```
        while (num != 0) {
```

```
            int digit = num % 10;
```

```
            reverse = reverse * 10 + digit;
```

```
            num = num / 10;
```

```
}
```

```
if (original == reverse) {
```

```
    System.out.println(original + " is a Palindrome number.");
```

```
} else {
```

```
    System.out.println(original + " is not a Palindrome number.");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter a number: 121

121 is a Palindrome number.

24. Write a Java program to find the maximum and minimum values in an integer array using a for-each loop.

```
import java.util.Scanner;
```

```
public class MaxMinArray {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the number of elements: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
System.out.println("Enter " + n + " elements:");  
for (int i = 0; i < n; i++) {  
    arr[i] = sc.nextInt();  
}  
  
int max = arr[0];  
int min = arr[0];  
  
// for-each loop  
for (int num : arr) {  
    if (num > max) {  
        max = num;  
    }  
  
    if (num < min) {  
        min = num;  
    }  
}  
  
System.out.println("Maximum value = " + max);  
System.out.println("Minimum value = " + min);  
  
sc.close();  
}  
}
```

OUTPUT:-

Enter the number of elements: 5

Enter 5 elements:

10 25 5 40 15

Maximum value = 40

Minimum value = 5

25. Create a Java program to calculate the average marks of students stored in a 2D array using a for-each loop.

```
import java.util.Scanner;
```

```
public class StudentAverage {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter number of students: ");  
        int students = sc.nextInt();  
  
        System.out.print("Enter number of subjects: ");  
        int subjects = sc.nextInt();  
  
        int[][] marks = new int[students][subjects];  
        int total = 0;  
        int count = 0;  
  
        System.out.println("Enter marks:");
```

```
for (int i = 0; i < students; i++) {  
    System.out.println("Student " + (i + 1) + ":");  
    for (int j = 0; j < subjects; j++) {  
        marks[i][j] = sc.nextInt();  
    }  
}  
  
// for-each loop  
for (int[] student : marks) {  
    for (int mark : student) {  
        total += mark;  
        count++;  
    }  
}  
  
double average = (double) total / count;  
  
System.out.println("Average marks = " + average);  
  
sc.close();  
}
```

OUTPUT:-

Enter number of students: 2

Enter number of subjects: 3

Enter marks:

Student 1:

80 75 90

Student 2:

70 85 80

Average marks = 80.0

26. Write a Java program that takes a day number (1–7) and displays whether it's a weekday or weekend using switch-case.

```
import java.util.Scanner;
```

```
public class DayType {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter day number (1-7): ");
```

```
        int day = sc.nextInt();
```

```
        switch (day) {
```

```
            case 1:
```

```
            case 2:
```

```
            case 3:
```

```
            case 4:
```

```
            case 5:
```

```
                System.out.println("Weekday");
```

```
                break;
```

```
            case 6:
```

case 7:

```
System.out.println("Weekend");
```

```
break;
```

default:

```
System.out.println("Invalid day number");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter day number (1-7): 6

Weekend

27. Create a Java program that takes a grade (A–F) and prints its qualitative meaning (e.g., A → Excellent, B → Good, etc.) using switch-case.

```
import java.util.Scanner;
```

```
public class GradeMeaning {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter grade (A-F): ");
```

```
        char grade = sc.next().toUpperCase().charAt(0);
```

```
        switch (grade) {
```

case 'A':

System.out.println("Excellent");

break;

case 'B':

System.out.println("Good");

break;

case 'C':

System.out.println("Average");

break;

case 'D':

System.out.println("Below Average");

break;

case 'E':

System.out.println("Pass");

break;

case 'F':

System.out.println("Fail");

break;

default:

System.out.println("Invalid grade");

```
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter grade (A-F): A

Excellent

28. Write a Java program to print numbers from 1 to 50, skipping all numbers that are perfect squares using continue.

```
public class SkipPerfectSquares {  
    public static void main(String[] args) {
```

```
        System.out.println("Numbers from 1 to 50 excluding perfect squares:");
```

```
        for (int i = 1; i <= 50; i++) {
```

```
            int root = (int) Math.sqrt(i);
```

```
            if (root * root == i) {
```

```
                continue;
```

```
            }
```

```
            System.out.print(i + " ");
```

```
        }
```

```
    }
```

```
}
```

OUTPUT:-

Numbers from 1 to 50 excluding perfect squares:

2 3 5 6 7 8 10 11 12 13 14 15 17 18 19 20 21 22 23 24 26 27 28 29 30 31 32 33 34 35 37 38 39
40 41 42 43 44 45 46 47 48 50

29. Create a Java program to generate random integers between 1 and 100 and stop when a number divisible by both 7 and 13 is found using break.

```
import java.util.Random;
```

```
public class RandomNumberBreak {
```

```
    public static void main(String[] args) {
```

```
        Random random = new Random();
```

```
        while (true) {
```

```
            int num = random.nextInt(100) + 1;
```

```
            System.out.println("Generated number: " + num);
```

```
            if (num % 7 == 0 && num % 13 == 0) {
```

```
                System.out.println("Number divisible by both 7 and 13: " + num);
```

```
                break;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

OUTPUT:-

Generated number: 45

Generated number: 82

Generated number: 19

Generated number: 91

Generated number: 26

Generated number: 14

Generated number: 98

Generated number: 91

Generated number: 84

Number divisible by both 7 and 13: 91

30. Write a Java program that:

❑ Uses shift operators to check if a number is a power of 4,

❑ Uses bitwise to toggle the 3rd bit of that number,

❑ Uses a for loop to print its multiplication table but skips multiples of 6 using continue,

❑ Stops if it reaches a multiple of 48 using break.

```
import java.util.Scanner;
```

```
public class BitwiseAndLoop {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
```

```
        int num = sc.nextInt();
```

```
        // Check if number is a power of 4 using shift operators
```

```
        boolean isPowerOf4 = false;
```

```
for (int i = 0; i < 16; i += 2) {  
    if ((1 << i) == num) {  
        isPowerOf4 = true;  
        break;  
    }  
}
```

```
if (isPowerOf4) {  
    System.out.println(num + " is a power of 4.");  
} else {  
    System.out.println(num + " is not a power of 4.");  
}
```

```
// Toggle the 3rd bit (bit position 2)
```

```
int toggled = num ^ (1 << 2);
```

```
System.out.println("After toggling 3rd bit = " + toggled);
```

```
// Multiplication table
```

```
System.out.println("Multiplication table:");
```

```
for (int i = 1; i <= 10; i++) {
```

```
    int result = num * i;
```

```
    // Skip multiples of 6
```

```
        if (result % 6 == 0) {  
            continue;  
        }  
  
        // Stop at multiples of 48  
        if (result % 48 == 0) {  
            break;  
        }  
  
        System.out.println(num + " x " + i + " = " + result);  
    }  
  
    sc.close();  
}  
}
```

OUTPUT:-

Enter a number: 4

4 is a power of 4.

After toggling 3rd bit = 0

Multiplication table:

4 x 1 = 4

4 x 2 = 8

4 x 4 = 16

4 x 5 = 20

4 x 7 = 28

4 x 8 = 32

$$4 \times 10 = 40$$